

ML Lab Week 13 Clustering Lab

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1. Based on the correlation heatmap and explained variance ratio from PCA, why was dimensionality reduction necessary for this dataset? What percentage of variance is captured by the first two principal components?

PCA was needed to reduce the 9-feature dataset (45,211 samples) to 2 dimensions for computational efficiency. The correlation matrix shows mostly weak correlations (maximum 0.17 between job and education), meaning features capture different information

The first two principal components capture 28.12% of total variance. This represents the maximum information we can preserve in 2D. The PCA scatter plot shows data concentrated in an elliptical cloud, revealing underlying structure. This reduction enables visual cluster interpretation while maintaining key data patterns.

2. Looking at both the elbow curve and silhouette scores, what is the optimal number of clusters for this dataset? Justify your answer using both metrics.

k=3 is optimal. The elbow plot shows inertia drops sharply from k=2 to k=3. After k=3, the decrease slows significantly. The curve clearly bends at k=3, showing diminishing returns beyond this point.

The silhouette score confirms this. K=3 achieves the highest score (0.3867), indicating the best cluster separation and cohesion. K=2 scores 0.3311 (too simple), while k=4 drops to 0.3581. Higher values (k=5-10) range 0.34-0.37 without improvement, suggesting over-segmentation.

3. Analyze the size distribution of clusters in both K-means and Bisecting K-means. Why do you think some clusters are larger than others? What might this tell us about the customer segments?

Standard K-means distribution:

- Cluster 0: 15,411 (34.1%)
- Cluster 1: 10,541 (23.3%)
- Cluster 2: 19,259 (42.6%)

Bisecting K-means distribution:

- Cluster 0: 20,434 (45.2%)
- Cluster 1: 16,348 (36.2%)
- Cluster 2: 8,429 (18.6%)

The algorithms produce different cluster sizes because they use different splitting strategies. Standard K-means finds all clusters simultaneously by optimizing global positions, creating a more balanced distribution (23-43% range). Bisecting K-means splits recursively, always dividing the largest cluster. The split tree shows it first divided all 45,211 samples into 20,434 and 24,777, then split the larger 24,777 group into 16,348 and 8,429. This hierarchical approach creates more imbalanced sizes (19-45% range).

Cluster 2 in standard K-means (42.6%) represents mainstream retail customers with typical banking behaviors. Most bank customers have similar patterns (moderate balances, standard products), naturally forming this large segment. The correlation matrix shows features vary independently (max correlation 0.17), letting customers group differently across dimensions.

Standard K-means produces a more interpretable distribution. No cluster falls below 23% or exceeds 43%. Bisecting K-means creates one dominant cluster (45.2%) and one small cluster (18.6%), making segment-specific strategies harder to implement. The PCA plot shows standard K-means clusters are spatially balanced, while bisecting K-means forces splits that may not match natural customer grouping.

4. Compare the silhouette scores between K-means and Recursive Bisecting K-means. Which algorithm performed better for this dataset and why do you think that is?

Standard K-means outperformed Bisecting K-means significantly:

- **Standard K-means:** Silhouette 0.387
- **Bisecting K-means:** Silhouette 0.338

Standard K-means achieved better results because it optimizes all three clusters simultaneously, finding the global best arrangement. It converged quickly in 27 iterations, showing the data has clear natural groupings.

Bisecting K-means scored lower because it makes decisions sequentially. The split tree reveals the process: first splitting all samples into 20,434 and 24,777, then dividing 24,777 into 16,348 and 8,429. Each split optimizes only that specific division without considering the final three-cluster structure.

5. Based on the clustering results in the PCA space, what insights can you draw about customer segmentation that might be valuable for the bank's marketing strategy?

Cluster 2 (42.6%, purple, right PCA region): This largest segment represents core retail customers with standard accounts. Located at positive PC1 values, they show higher banking engagement or established relationships.

Cluster 0 (34.1%, turquoise, left PCA region): This segment achieved the highest silhouette score (median ~0.6), meaning highly similar customers. Positioned at negative PC1 and positive PC2 values (0 to 8), they likely have lower balances or fewer products.

Cluster 1 (23.3%, yellow, center PCA region): The smallest cluster with widest silhouette distribution suggests diverse customer profiles. Located centrally with moderate PC values, they may hold multiple products or represent transition states. This heterogeneity requires personalized approaches.

6) In the PCA scatter plot, we see three distinct colored regions (turquoise, yellow, and purple). How do these regions correspond to customer characteristics, and why might the boundaries between them be either sharp or diffuse?

The PCA scatter plot displays three spatial regions created by standard K-means. PC1 (14.88% variance) represents the primary dimension, likely banking engagement, account maturity, or balance level. The negative-to-positive axis shows a progression from basic to complex banking relationships. PC2 (13.24% variance) captures secondary variation, possibly demographics (age), employment stability, or specific product holdings like loans. The vertical spread shows how customers vary independently of PC1 characteristics.

Cluster boundaries show different sharpness levels. The Cluster 0/Cluster 2 boundary is sharp with minimal overlap. Cluster 0's high silhouette coefficient (median ~0.6) confirms this clear separation. This sharp divide suggests the underlying features represent categorical differences.

The Cluster 1/Cluster 2 boundary is diffuse. Customers here exist on continuums rather than discrete categories. Cluster 1's wider silhouette distribution (box plot shows larger spread) reflects this ambiguity. Some points are confidently assigned while others sit near decision boundaries.



